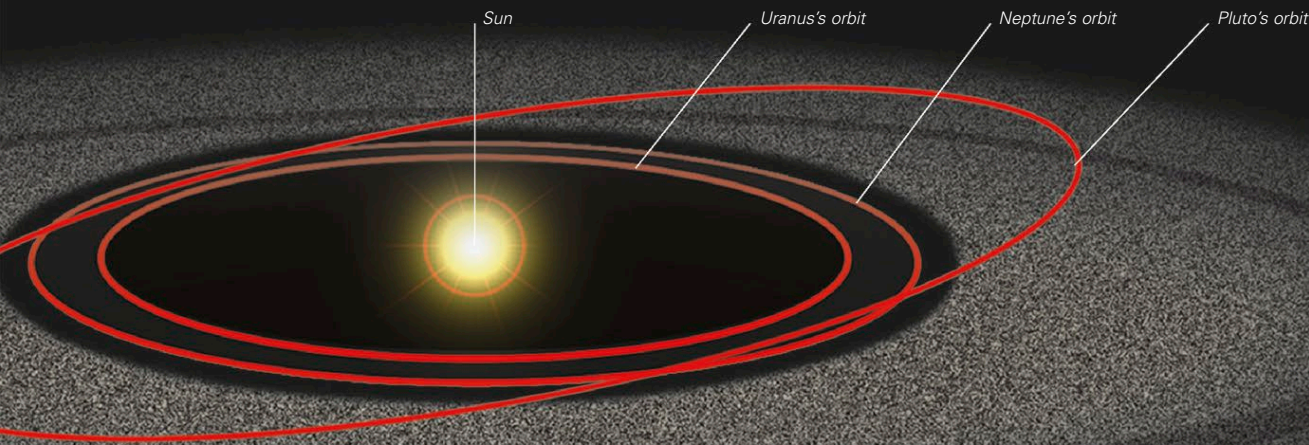




THE KUIPER BELT AND ITS CONSTITUENTS

The Kuiper Belt is a broad ring of objects that begins around the orbit of Neptune and extends out to roughly 9.3 billion miles (15 billion km) from the Sun. The possibility of such a belt was initially put forward in 1930, soon after the discovery of Pluto (see pp.208–209). The first theoretical models for how such a belt could have formed were proposed in 1943 by British astronomer Kenneth Edgeworth and in 1951 by Gerard Kuiper (see opposite). For this reason, the belt is sometimes known as the Edgeworth–Kuiper Belt, or EKB. However, the belt remained purely theoretical until 1992, when astronomers identified a small body with a diameter of about 100 miles (160 km), now known as 1992 QB1. This was the first confirmation that there were other objects in addition to Pluto in the space beyond Neptune, and since then, about a thousand more such objects have been discovered. The Kuiper Belt as a whole can be split into an inner zone called the Classical Kuiper Belt and an outer zone called the Scattered Disk. The Classical Kuiper Belt extends out to about 4.7 billion miles (7.5 billion km) from the Sun and is relatively densely populated with objects that have roughly circular orbits. The drop in density at its outer edge is known as the Kuiper Cliff. Beyond this is the Scattered Disk, which is relatively sparsely populated with objects that have more eccentric and tilted orbits.



ARROKOTH

The only small Kuiper Belt object to have been seen at close range, this 21-mile (36-km) long object consists of two round, icy objects that must have become stuck together after a low-speed collision. Such a body is referred to as a “contact binary.”

EXPLORING SPACE

NEW HORIZONS

This NASA mission was the first (and so far only) spacecraft to visit any object more remote than Neptune. It was launched in January 2006, and flew past Pluto and its moons in July 2015. Its subsequent trajectory was adjusted to pass close to a newly discovered small Kuiper Belt Object temporarily designated as 2014MU 69, which later received the permanent name 486958 Arrokoth. New Horizons made a successful flyby of this on January 1, 2019, passing within 2,200 miles (3,500 km) at a speed of 32,000 mph (51,500 kph). The spacecraft remains healthy, and its radioactive power source (consisting of plutonium-238) should last well into the 2030s. The search is on to locate an even more distant KBO that is close enough to New Horizons' trajectory that there is enough fuel remaining to adjust the spacecraft's course to achieve another close flyby.

